PUTTING ALL MARKETS TOGETHER: THE AS-AD MODEL

MACROECONOMICS

PAULINA ETXEBERRIA GARAIGORTA

Reinventing Higher Education

OUTLINE

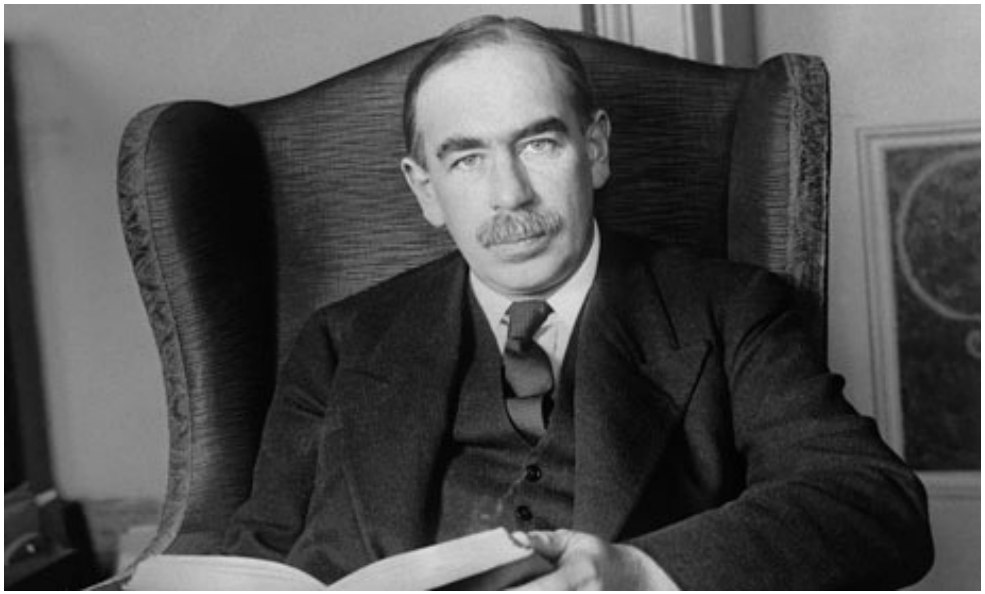
1. The basic model of Economic fluctuations:

   The AD-AS Model

2. AS in the medium/long-run: LRAS curve
3. AS in the short-run: SRAS curve
4. AD: AD Curve
5. The AD-AS model: Equilibrium in the short & long-run
6. Causes of Economic Fluctuations
7. Stabilization Policy
8. Economic Shocks & Policies together

1. The Basic Model of Economic Fluctuations: The AD-AS Model

The Origins of the Model of AD and AS:
John Maynard Keynes, 1883–1946



Recessions and depressions

can result from **Inadequate Demand**;
policymakers should shift *AD*.

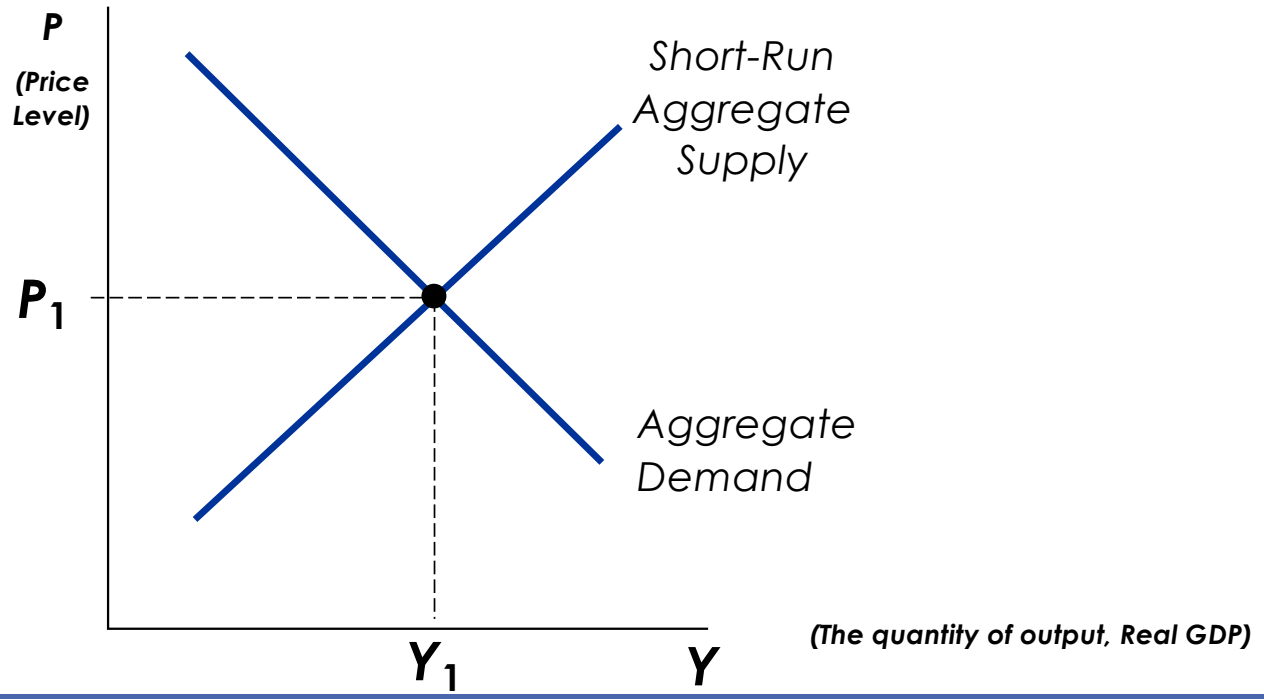
The General Theory of Employment, Interest, and Money, 1936

1. The Basic Model of Economic Fluctuations: The AD-AS Model

The model that most economist use to explain short-run fluctuations in economic activity around its long-run trend.

The model determines

- *the equilibrium price level (CPI or GDP deflator) and*
- *- the equilibrium output (real GDP).*



1. The Basic Model of Economic Fluctuations: The AD-AS Model

The model assumes equilibrium in the three markets:

1. Goods & Services market (IS curve)
2. Financial market (LM; Policy rule for Central Bank)

} AD Curve

3. Labor market (Phillips curve) → AS curve

1. The Basic Model of Economic Fluctuations: The AD-AS Model

1. IS curve: $\tilde{Y}_t = \bar{a} - \bar{b}(R_t - \bar{r})$

2. Policy rule: $R_t - \bar{r} = \bar{m}(\pi_t - \bar{\pi})$

Substitute the policy rule into the IS curve to get the AD curve

AD curve: $\tilde{Y}_t = \bar{a} - \bar{b}\bar{m}(\pi_t - \bar{\pi})$

Short-run output is a function of the inflation rate

AS curve: $\pi_t = \pi_{t-1} + \bar{\nu}\tilde{Y}_t + \bar{o}$

EXTRA

1. The Basic Model of Economic Fluctuations: : The AD-AS Model

How the Short Run Differs from the Medium/Long Run

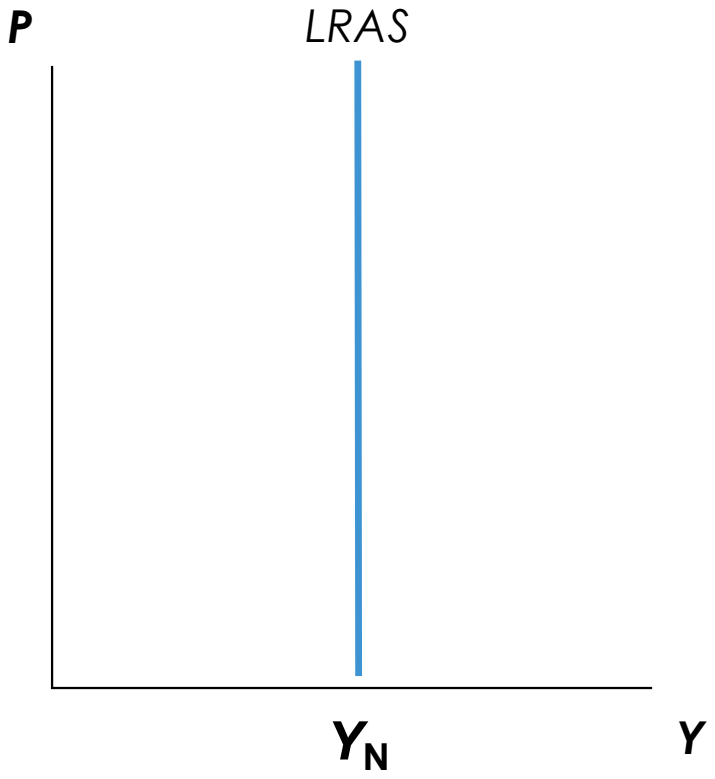
- Most economists believe that

  Classical Theory describes the world in the long run but not in the short run.

  - Changes in the money supply affect nominal variables but not real variables in the long run.
  - The assumption of **monetary neutrality** is not appropriate when studying year-to-year changes in the economy.

2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

The long-run aggregate-supply curve (LRAS) is VERTICAL at the **natural rate of output =** potential output = full-employment output.



2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

In the medium/long run, output is determined by $\bar{Y} = A\,F(K, H, N)$

- A: Technological Progress ← Long run
- K: Physical Capital K
- H: Human Capital H = Labor L and Education e ← Medium run
- N: Natural resources

$\bar{Y}$ or Y_N is the **full-employment, natural** level of output, **potential** output, the level of output at which the economy's resources are fully employed.

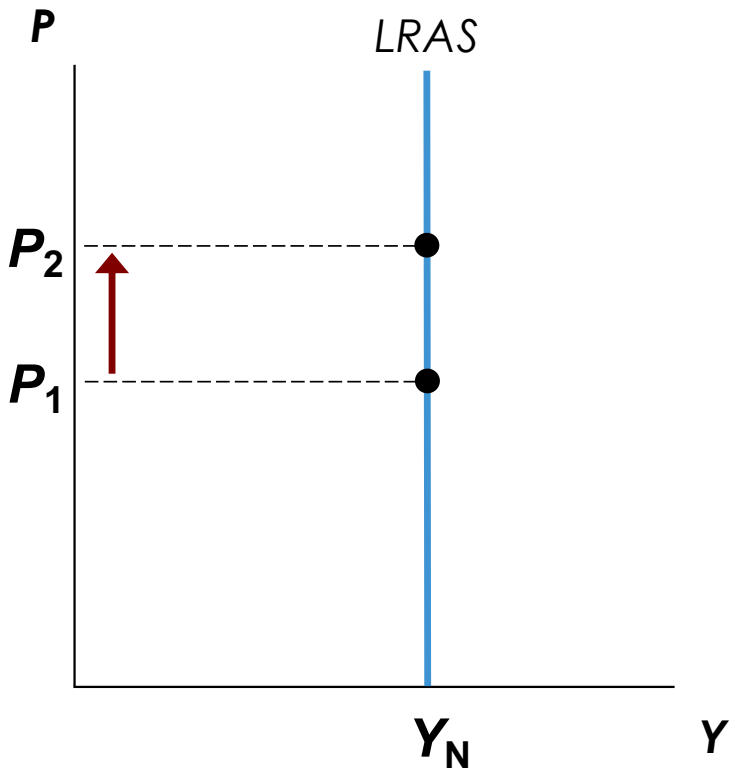
"Full employment" means that unemployment equals its natural rate (not zero).

2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

Y_N determined by the economy's stocks of labor, capital, and natural resources, and on the level of technology.

An increase in ***P***

does not affect any of these, so it does not affect $\boldsymbol{Y_N}$.



2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

Why the Medium/Long-Run Aggregate-Supply Curve Might Shift?

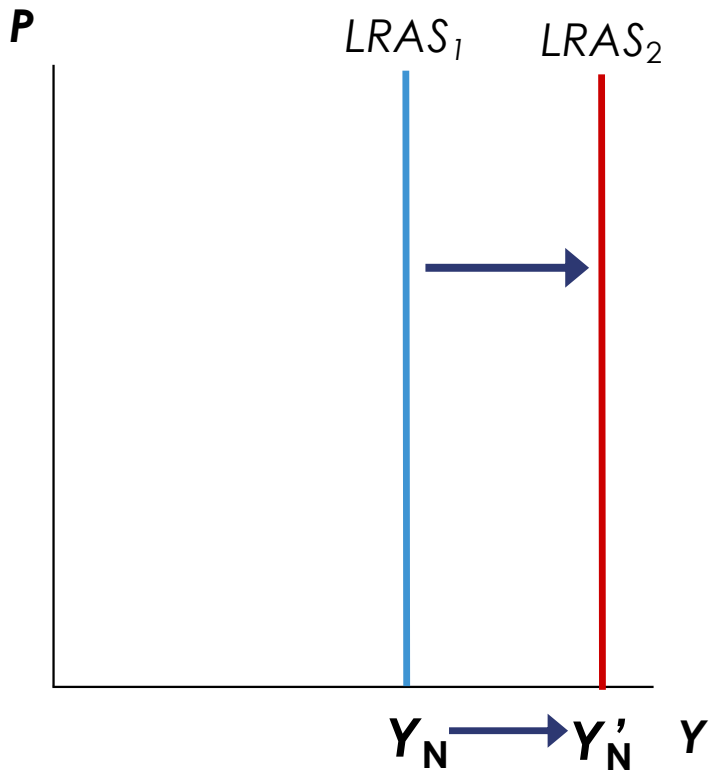
Any change in the economy that alters the natural rate of output shifts the medium/long-run aggregate supply curve

$$\bar{Y} = A\,F(K, H, N)$$

The shifts may be categorized according to the various factors in the classical model that affect output.

2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

Example: Immigration increases *L*, causing Y_N to rise.



2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

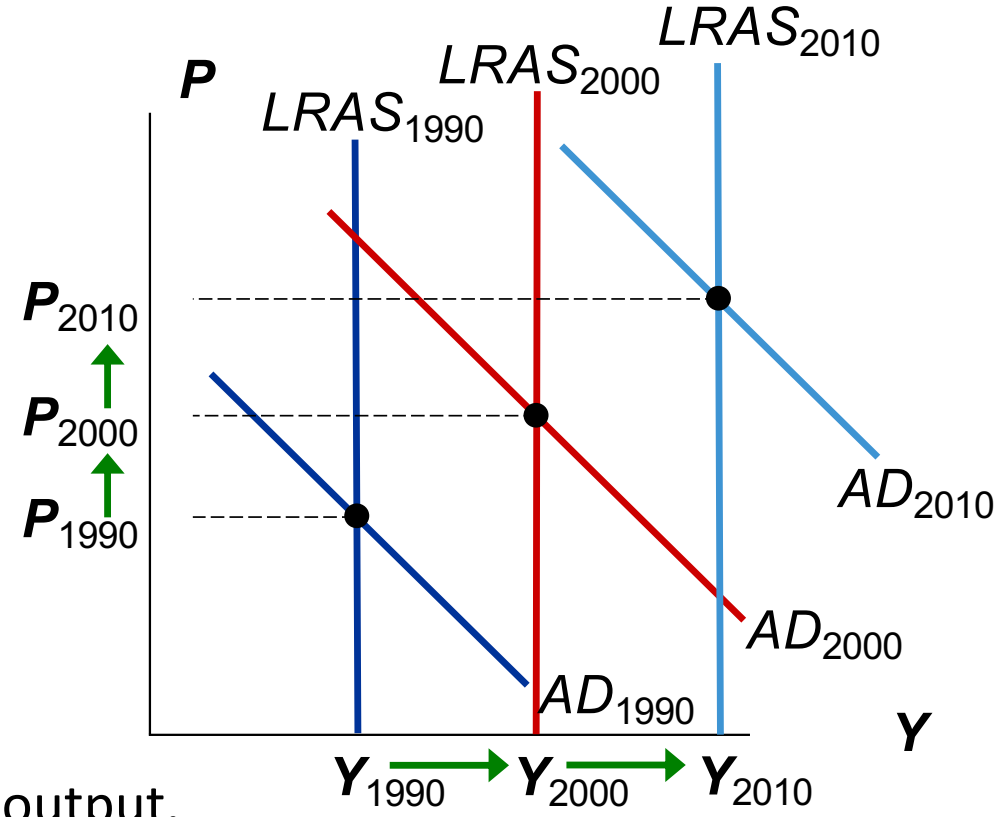
1. **Changes in Labor *L* natural rate of unemployment:** Immigration, Baby-boomers retire, Government policies that change labor market regulations, change in minimum wage, education, more people get college degree, labor productivity
2. **Changes in *K*:** Investment in factories, Equipment, Factories destroyed by a hurricane
3. **Changes in *N***: Discovery of new mineral deposits, Reduction in supply of imported oil, Changing weather patterns that affect agricultural production
4. **Changes in *A*:** Productivity improvements from technological progress, invention of the computer, innovations, institutions

2. Aggregate Supply in the MEDIUM/LONG-RUN: LRAS Curve

Using Aggregate Demand and Aggregate Supply to Depict Medium/Long-Run Growth and Inflation

Over the medium/long run, tech. progress shifts *LRAS* to the right

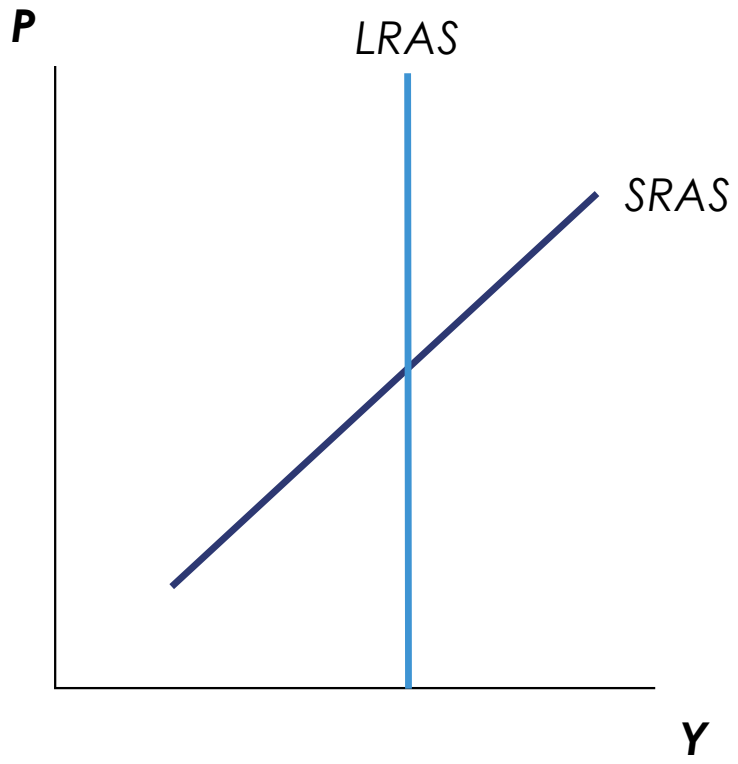
and growth in the money supply shifts *AD* to the right.



Result:
ongoing inflation and growth in output.

3. Aggregate Supply in the SHORT-RUN: SRAS Curve

The ***AS* curve** shows the total quantity of goods and services firms produce and sell at any given price level.



AS is:

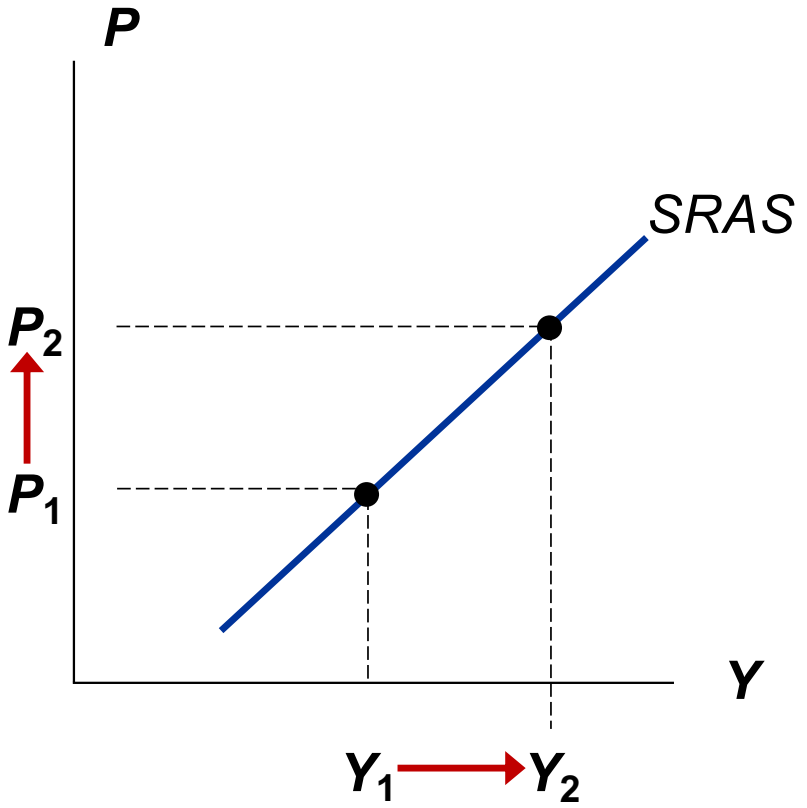
- vertical in long run
- upward-sloping in short run

3. Aggregate Supply in the SHORT-RUN : SRAS Curve

The *SRAS* curve
is upward sloping:

Over the period
of 1-2 years,
an increase in ***P***

causes an increase in the
quantity of goods and
services supplied.



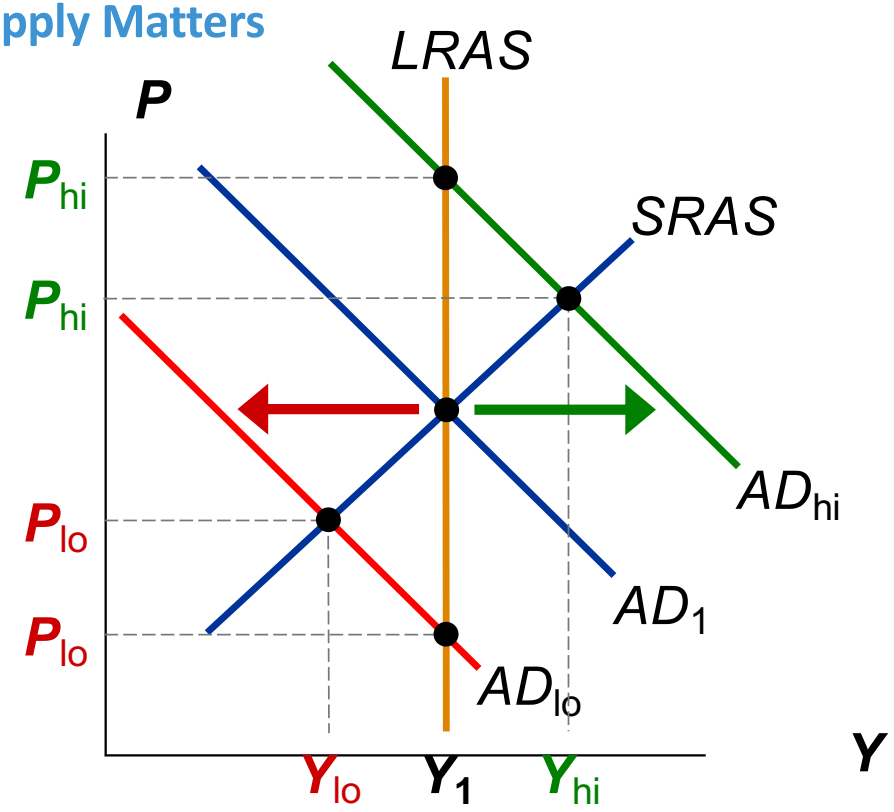
The positive slope of the SRAS is the key to understanding short-run fluctuations.

3. Aggregate Supply in the SHORT-RUN : SRAS Curve

Why the Slope of Short Run Aggregate Supply Matters

- If *AS* is vertical, fluctuations in *AD* do not cause fluctuations in output or employment.

- If *AS* slopes up, then shifts in *AD* do affect output and employment.



3. Aggregate Supply in the SHORT-RUN: SRAS Curve

Short-run fluctuations in output and price level should be viewed as deviations from the continuing long-run trends.

-In the short run, an increase in the overall level of prices in the economy tends to raise the quantity of goods and services supplied.

- A decrease in the level of prices tends to reduce the quantity of goods and services supplied.

3. Aggregate Supply in the SHORT-RUN : SRAS Curve

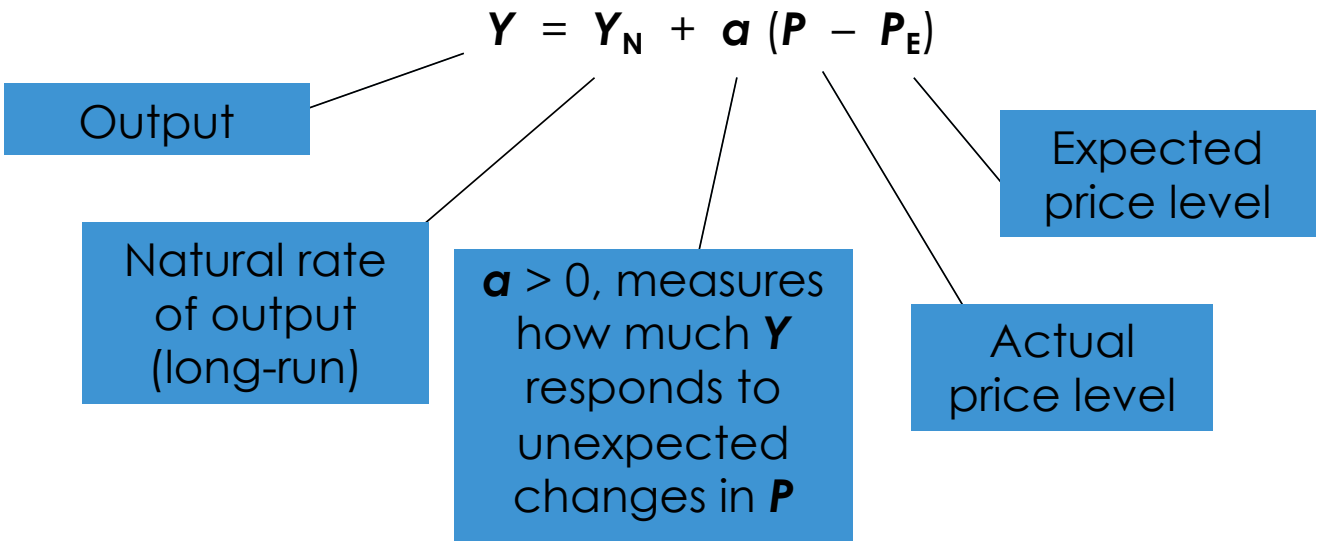
Three Theories of Short Run Aggregate Supply Curve

1. **The Sticky-Wage Theory**
2. **The Sticky-Price Theory**
3. **The Misperceptions Theory**

3. Aggregate Supply in the SHORT-RUN : SRAS Curve

What the 3 Theories Have in Common:

In all 3 theories, Y deviates from Y_N when P deviates from P_E.



3. Aggregate Supply in the SHORT-RUN : SRAS Curve

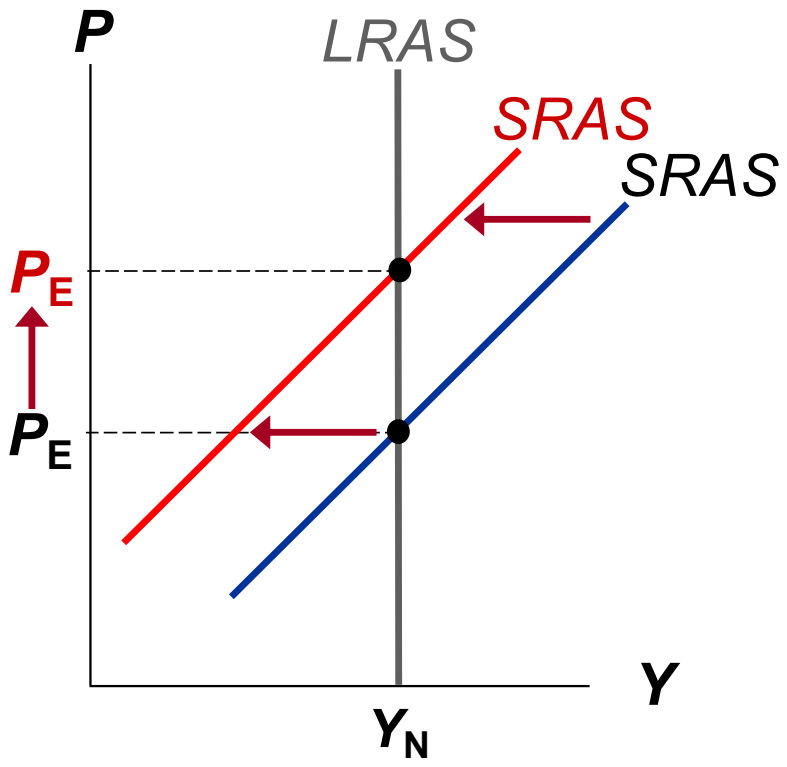
Why the Short Run Aggregate Supply Curve Might Shift

- Everything that shifts *LRAS* shifts *SRAS*, too.

- Also, P_E shifts *SRAS*:

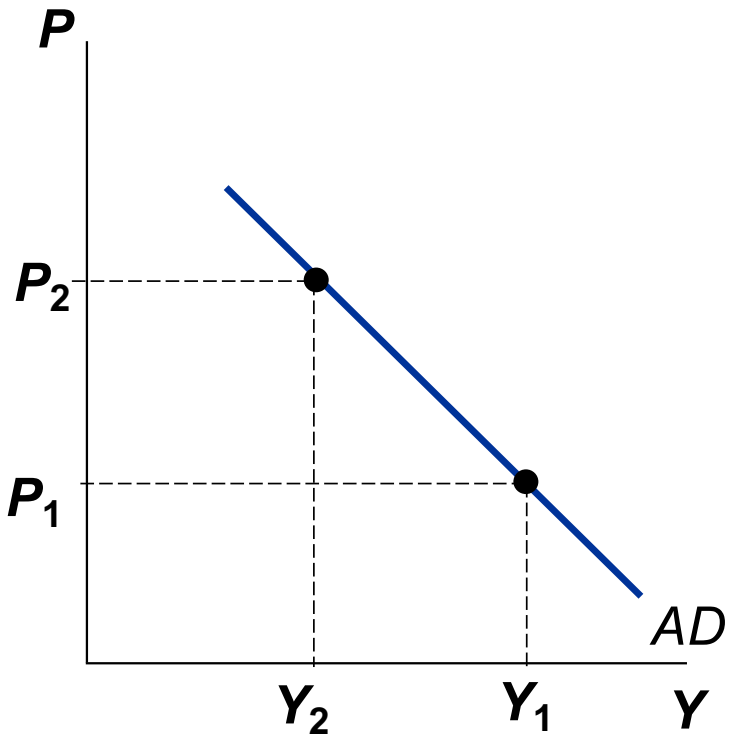
If P_E rises, workers & firms set higher wages.

At each ***P***, production is less profitable, ***Y*** falls, *SRAS* shifts left.



4. Aggregate Demand: AD Curve

The AD curve shows
the quantity of
all good and services
demanded
in the economy at any
given price level.



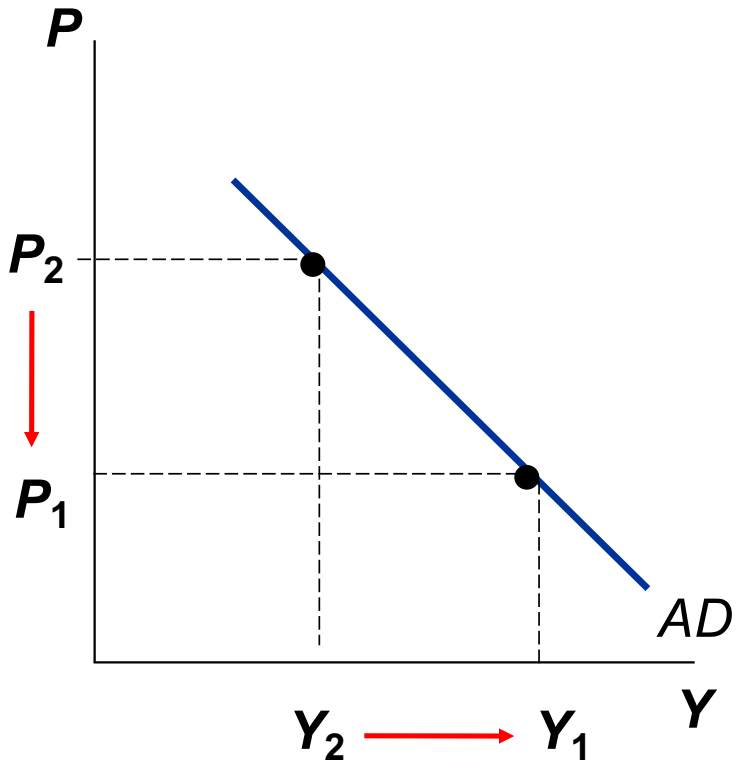
4. Aggregate Demand: AD Curve

Why the Aggregate Demand Curve Slopes Downward

The four components of GDP (*Y*) contribute to the AD for goods & services.

$$Y = C + I + G + NX$$

Y = real GDP
C = the sum of its consumption
I = Investment
G= Government Purchases
NX = Net exports



Assume G fixed by government policy.

To understand the slope of AD, must determine how a change in P affects C, I, and NX.

4. Aggregate Demand: AD Curve

Why the Aggregate Demand Curve Might Shift

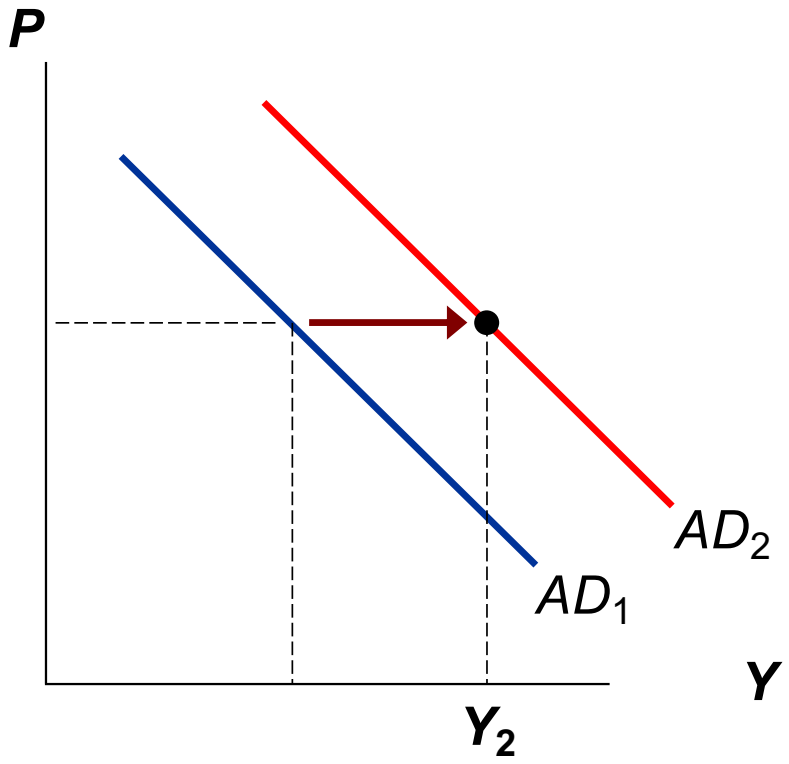
Any event that changes ***C***, ***I***, ***G***, or ***NX***

—except a change in ***P***—

will shift the *AD* curve.

Example:

A stock market boom makes households feel wealthier, ***C*** rises, the *AD* curve shifts right.



4. Aggregate Demand: AD Curve

Shifts Arising from Changes in Demand Components

- Changes in ***C:*** Stock market boom/crash; Preferences; consumption/saving tradeoff; Tax hikes/cuts

- Changes in ***I:*** Firms buy new computers, equipment, factories; Expectations, optimism/pessimism; Interest rates, monetary policy; Investment Tax Credit or other tax incentives

- Changes in ***G:*** Federal spending, e.g., defense; State & local spending, e.g., roads, schools

- Changes in ***NX:*** Booms/recessions in countries that buy our exports; Appreciation/depreciation resulting from international speculation in foreign exchange market.

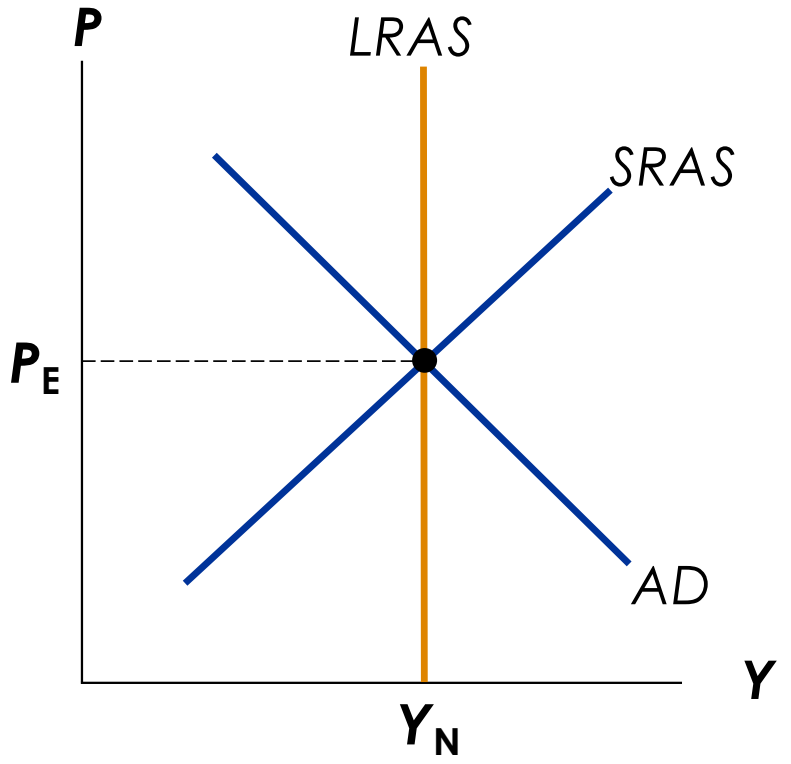
5. The AS-AD Model: Equilibrium in the short & medium/long-run

In the medium/long-run equilibrium,

$P_E = P$,

$Y = Y_N$,

and unemployment is at its natural rate.



6. Causes of Economic Fluctuations

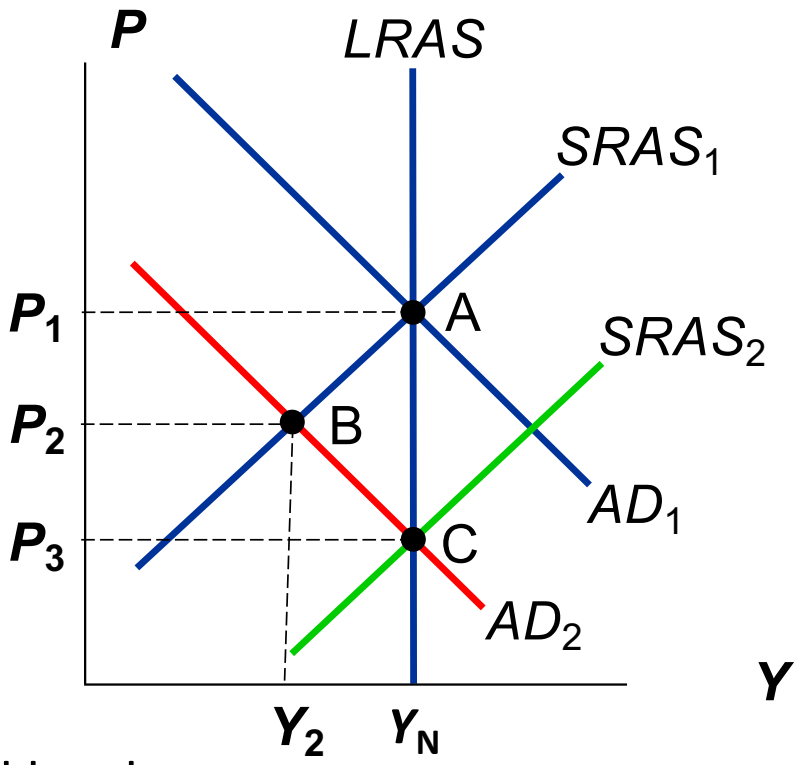
- Caused by events that shift the *AD* and/or *AS* curves.
- **Shocks**: exogenous changes in aggregate supply or demand
- Shocks temporarily push the economy away from full employment.
- Four steps to analyzing economic fluctuations:
  1. Determine whether the event shifts *AD* or *AS*.
  2. Determine whether curve shifts left or right.
  3. Use *AD*–*AS* diagram to see how the shift changes ***Y*** and ***P*** in the short run.
  4. Use *AD*–*AS* diagram to see how economy moves from new SR eq'm to new LR eq'm.

6. Causes of Economic Fluctuations

1. The Effects of a Shift in Aggregate Demand (AD)

Event: Stock market crash

1. Affects ***C***, *AD* curve
2. ***C*** falls, so *AD* shifts left
3. SR equilibrium at B.
   P and ***Y*** lower,
   unemployment higher
4. Over time, P_E falls,
   SRAS shifts right,
   until LR equilibrium at C.
   Y and unemployment back at initial levels.



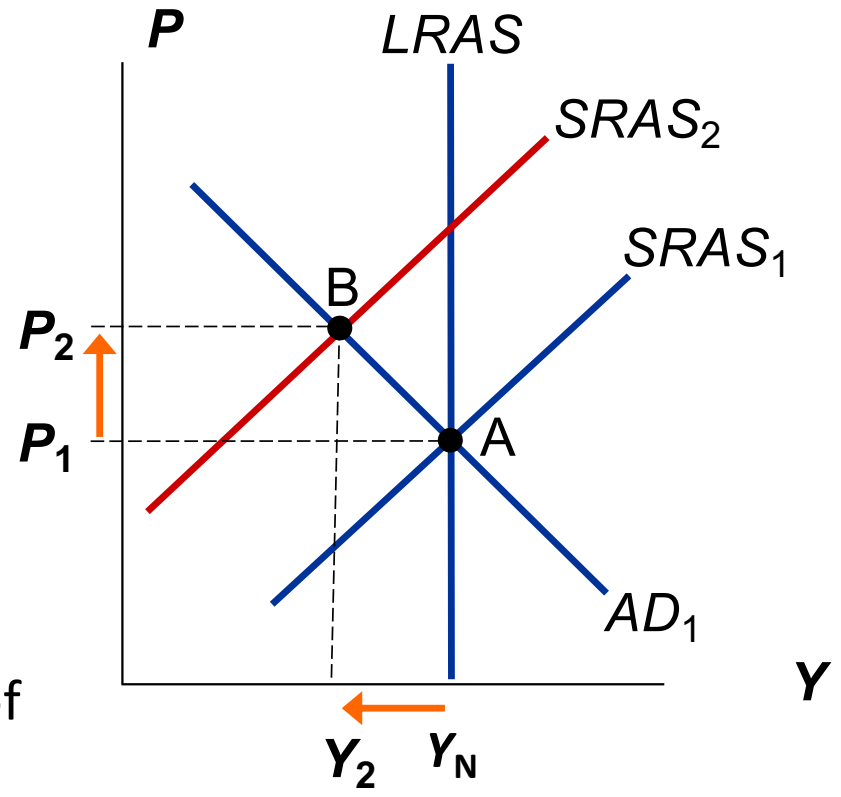
6. Causes of Economic Fluctuations

2. The Effects of a Shift in Aggregate Supply (AS)

Event: Oil prices rise (temporary)

1. Increases costs, shifts *SRAS* *(assume LRAS constant)*
2. *SRAS* shifts left
3. SR eq'm at point B. ***P*** higher, ***Y*** lower, unemp higher

   From A to B, **stagflation**, a period of falling output and rising prices.

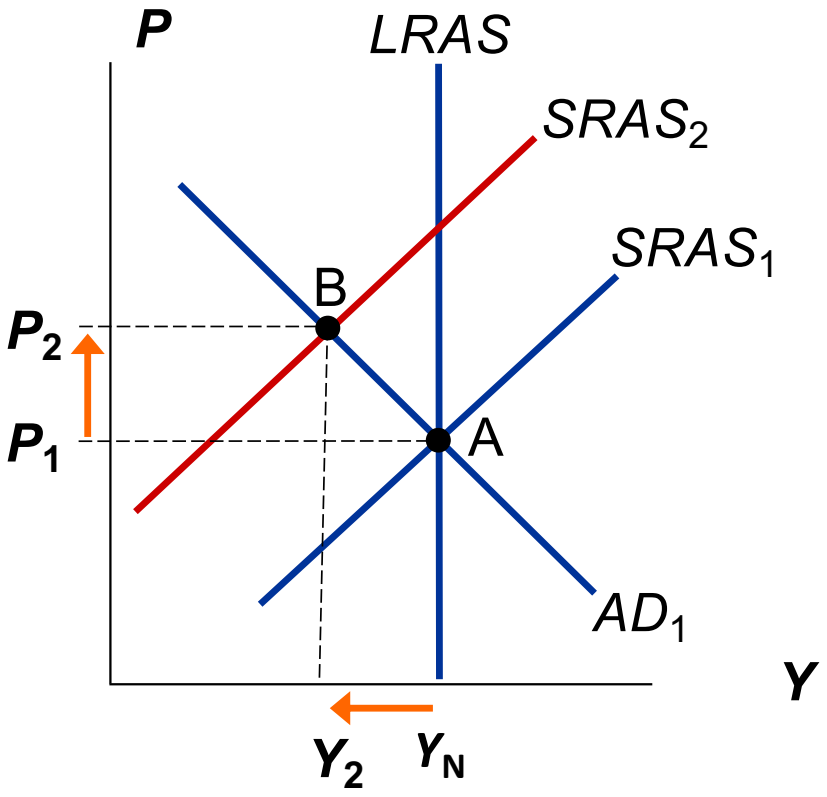


6. Causes of Economic Fluctuations

2. The Effects of a Shift in Aggregate Supply (AS)

Event: Oil prices rise

4. In the Medium/Long run, prices will decrease and the eq. will move back to A in the medium/long-run.



7. Stabilization Policy

Definition: *policy actions aimed at reducing the severity of short-run economic fluctuations (severity of the business cycle).*

FISCAL & MONETARY POLICY  **SHIFTS IN AD**

Example: Using monetary policy to combat the effects of adverse supply shocks.

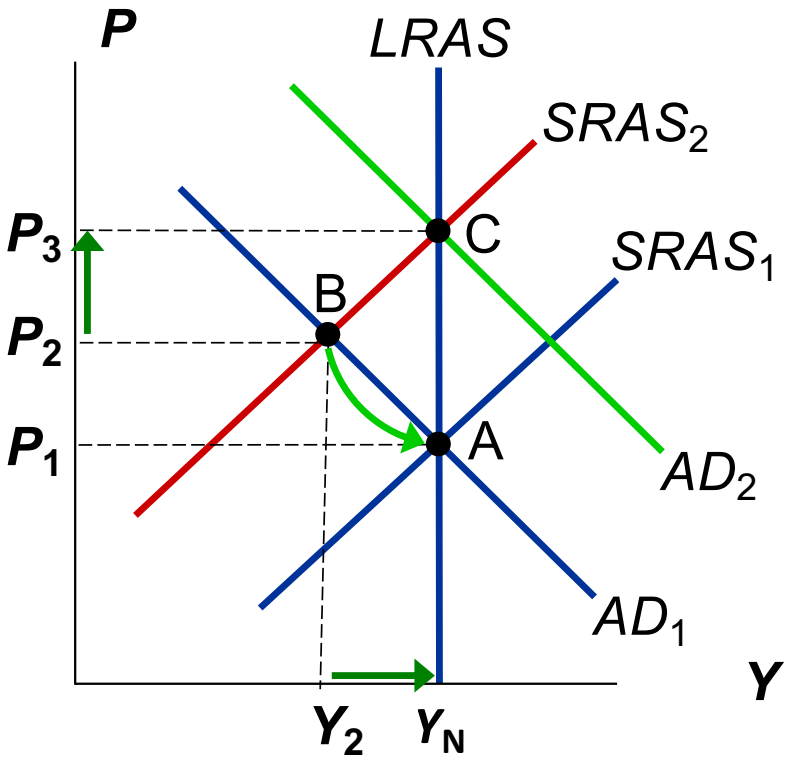
7. Stabilization Policy

Accommodating an Adverse Shift in Aggregate Supply (AS)

Event: Oil prices rise

Faced with an adverse shift in aggregate supply from AS1 to AS2, policymakers who can influence aggregate demand might try to shift the aggregate-demand curve to the right from AD1 to AD2. The Economy would move from point A to point C.

This policy would prevent the supply shift from reducing output in the short run, but the price level would permanently rise from P1 to P3.



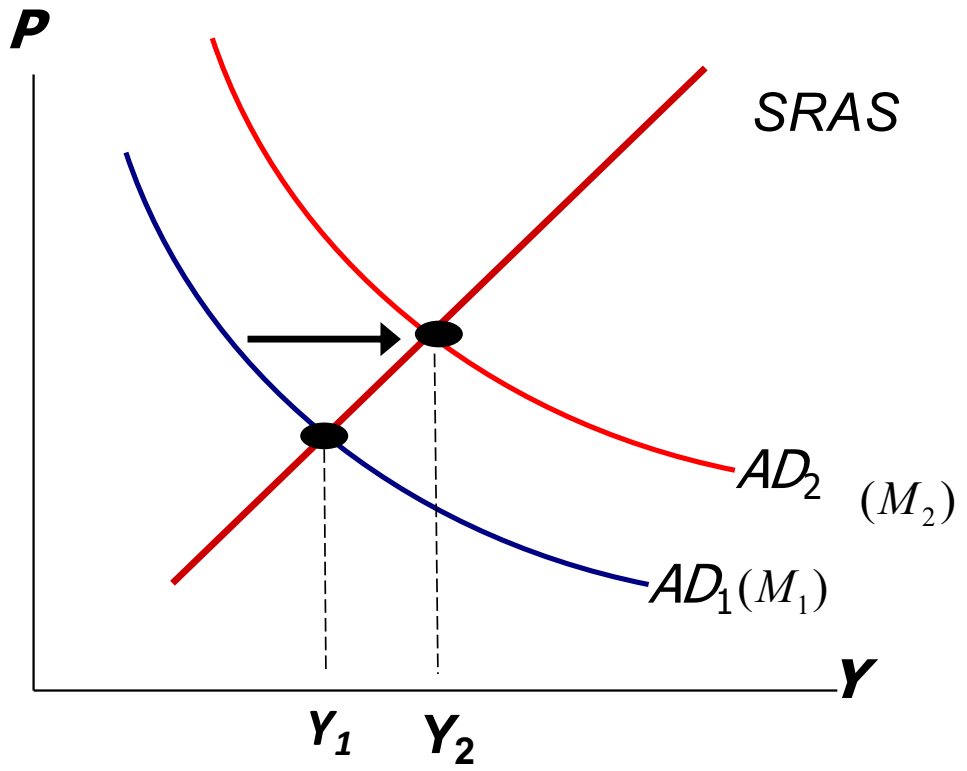
7. Stabilization Policy

An increase in the money supply shifts the *AD* curve to the right.

$$M_2 > M_1$$

An Expansionary Monetary Policy

Shifting the *AD* curve

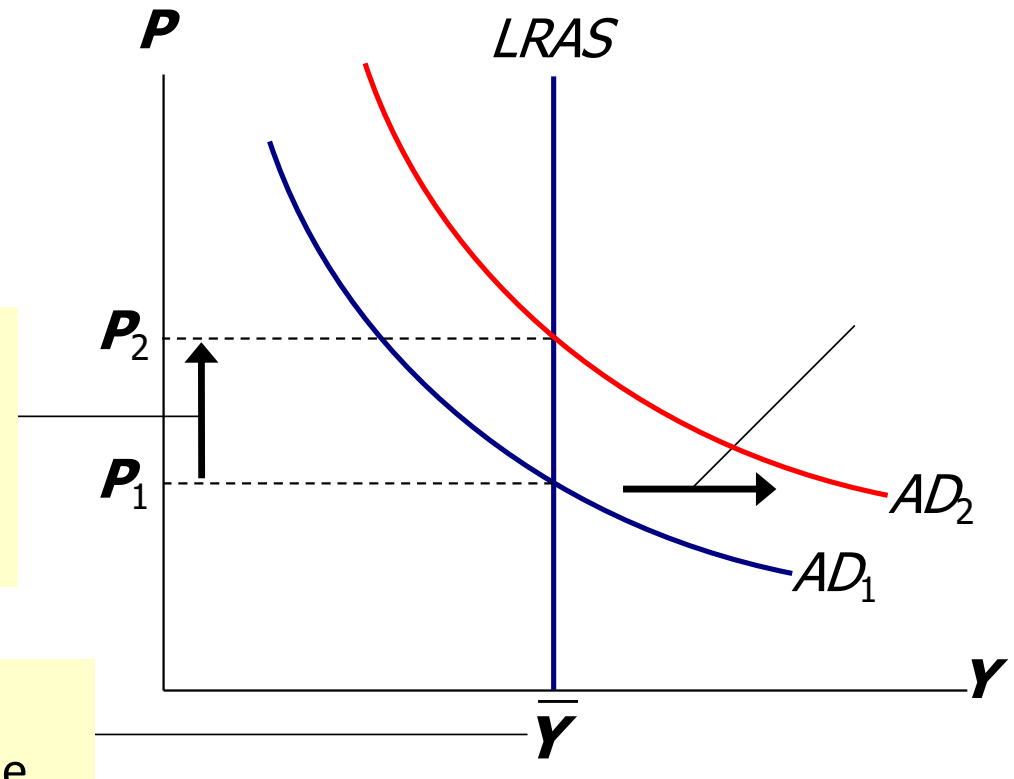


Note: The same would occur with an Expansionary Fiscal Policy (G, T).

7. Stabilization Policy

An Expansionary Monetary Policy

Medium/Long-run effects of an increase in *M*



An increase in ***M*** shifts *AD* to the right.

In the medium/long run, this raises the price level…

…but leaves output the same.

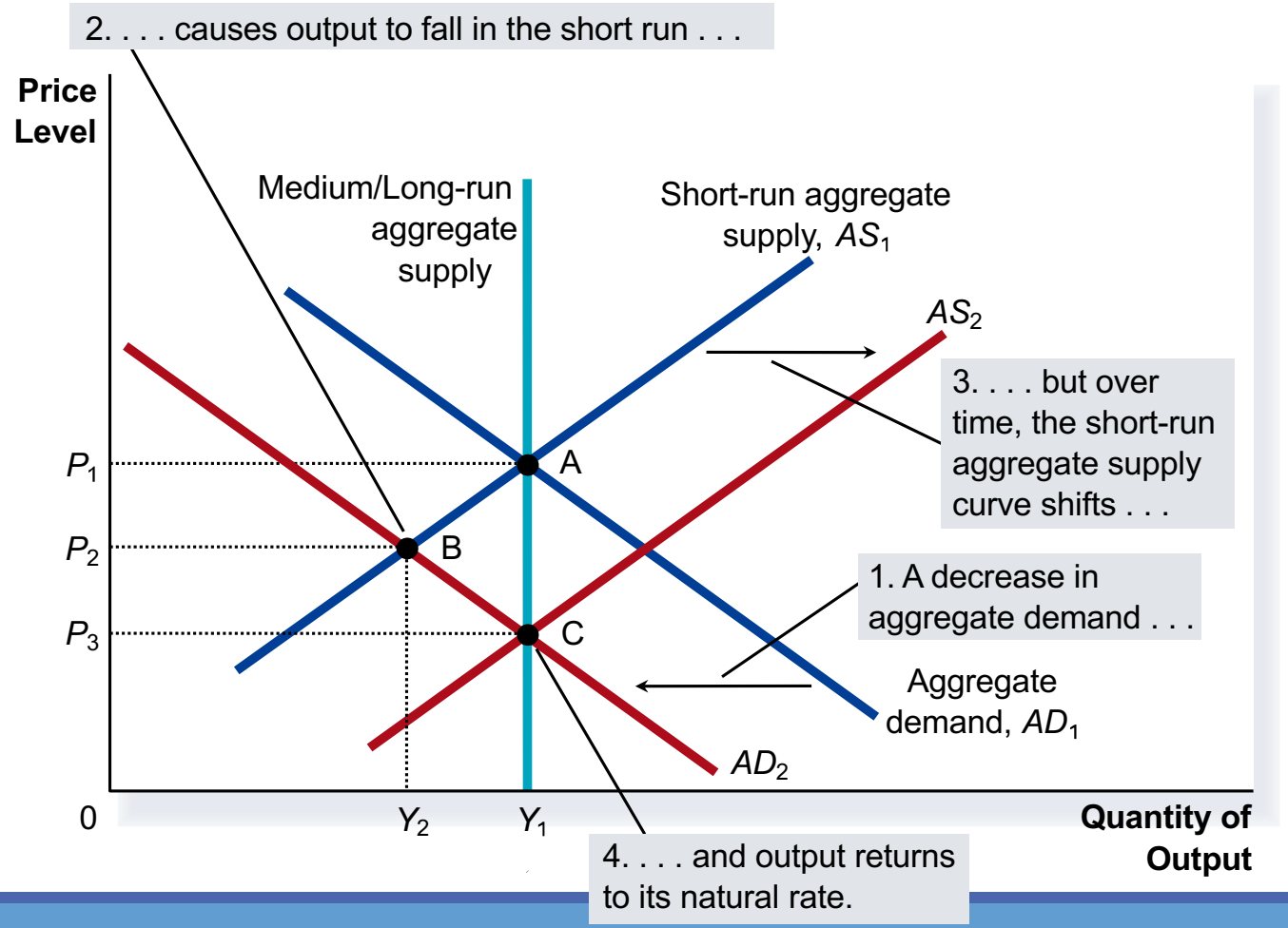
8. Economic Shocks & Policies together

Exercise 1:

Imagine that the economy is in its medium/long-run equilibrium and there are Expectations about the economy getting into a recession and, as a consequence, people consume less.

1. In our AD & AS Model, it can be represented as a movement of the AS curve to the left in the short-run. True or False?
2. How is the dynamic movement to the long-run equilibrium?
3. Which Stabilization Policy could be applied in the short-run?

Exercise 1: A Contraction in Aggregate Demand



Exercise 1: A Contraction in Aggregate Demand

1. False. It represent a Shifts in Aggregate Demand to the left.

2. In the short run, shifts in AD cause fluctuations in the economy's output of goods and services.

In the medium/long run, shifts in AD affect the overall price level but do not affect output.

3. Expansionary Fiscal or Monetary Policy.

8. Economic Shocks & Policies together

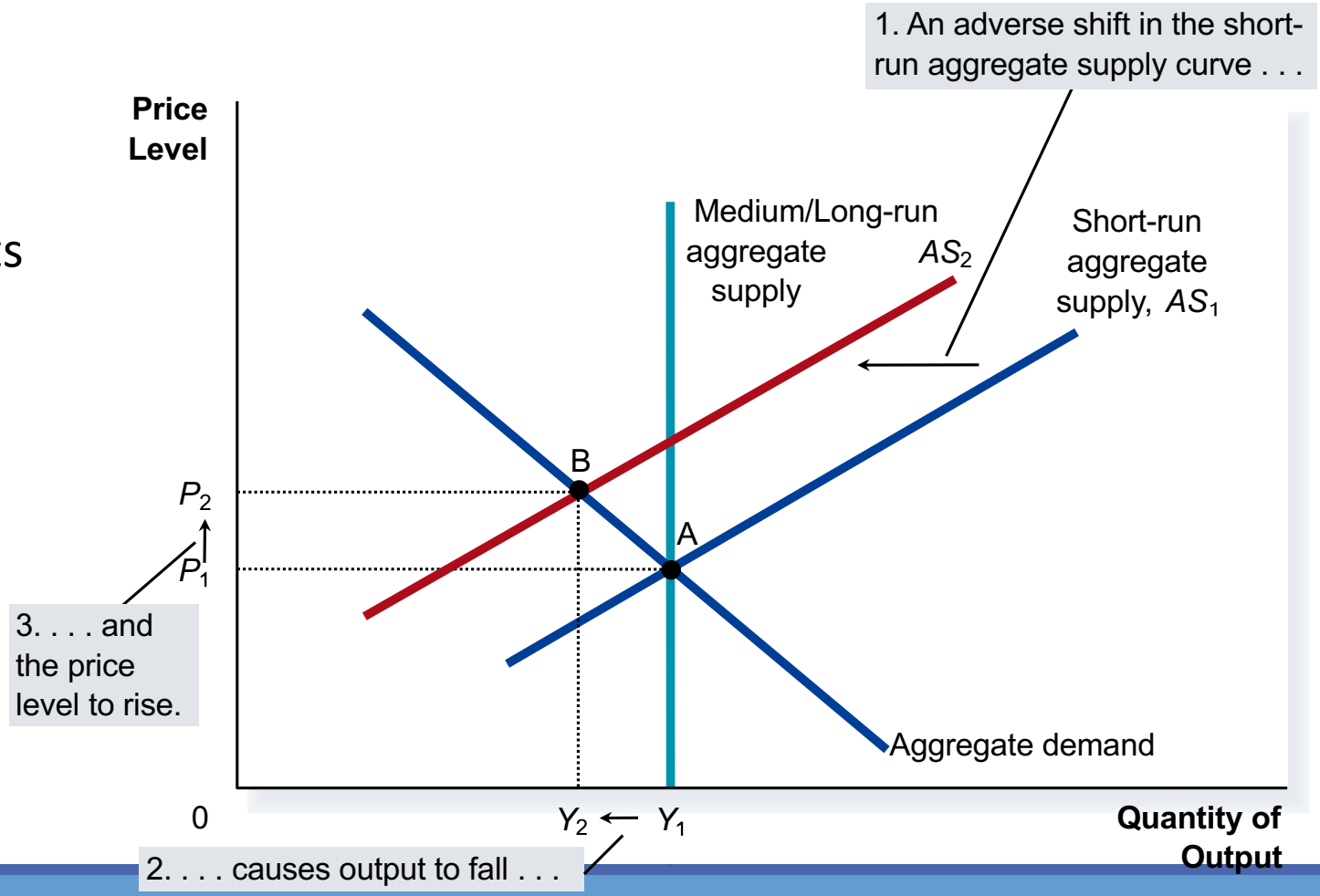
Exercise 2:

Imagine that the economy is in its medium/long-run equilibrium and the Government decides to increase social security taxes paid by firms to cover for the Pension System in the short-run.

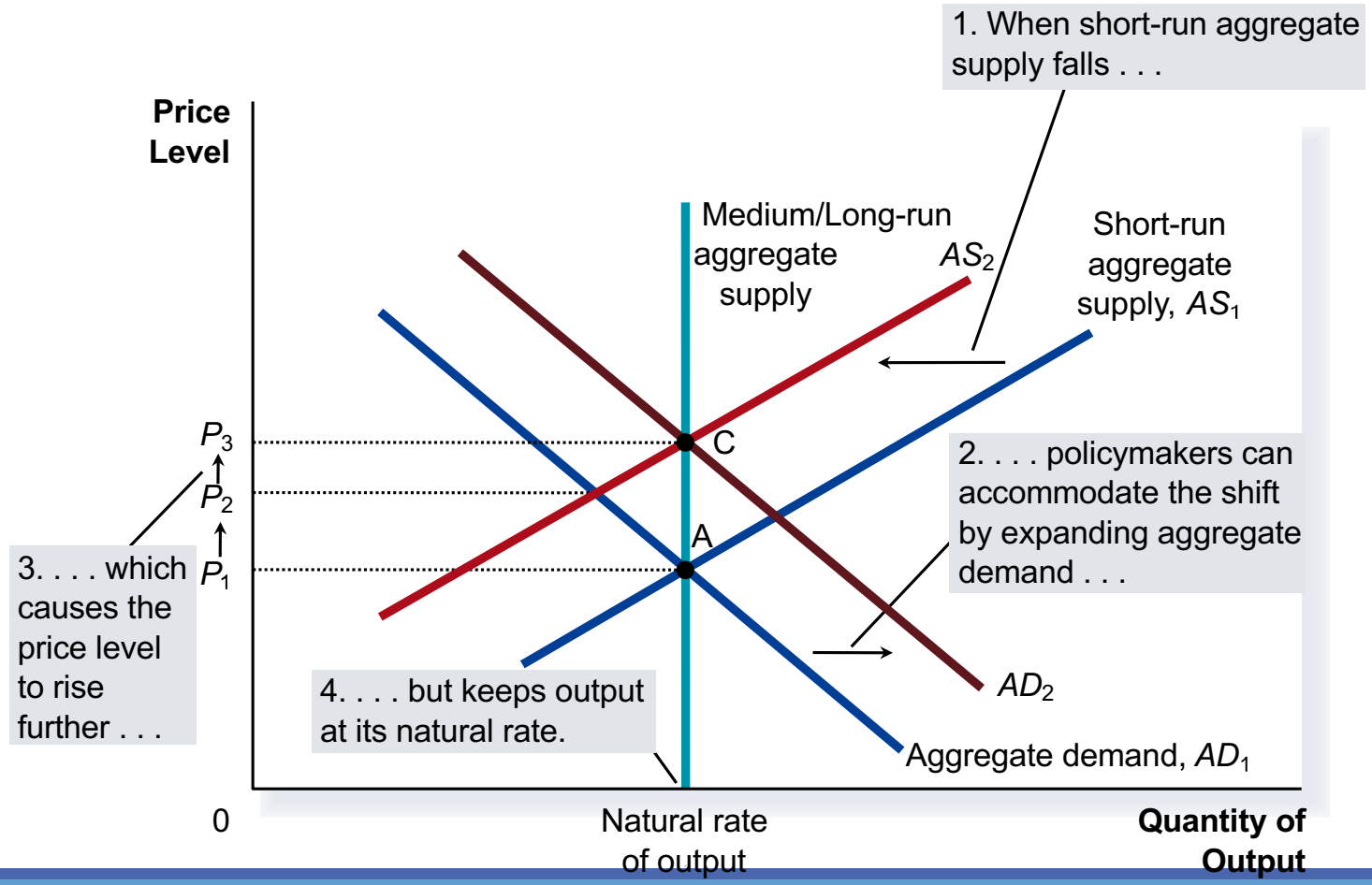
1. In our AD & AS Model, it can be represented as a movement of the AS curve to the left in the short-run. True or False?
2. How is the dynamic movement to the medium/long-run equilibrium?
3. Which Stabilization Policy could be applied in the short-run?

Exercise 2: An Adverse Shift in Aggregate Supply

1. True.
Labor costs increase.



Exercise 2: Accommodating an Adverse Shift in Aggregate Supply



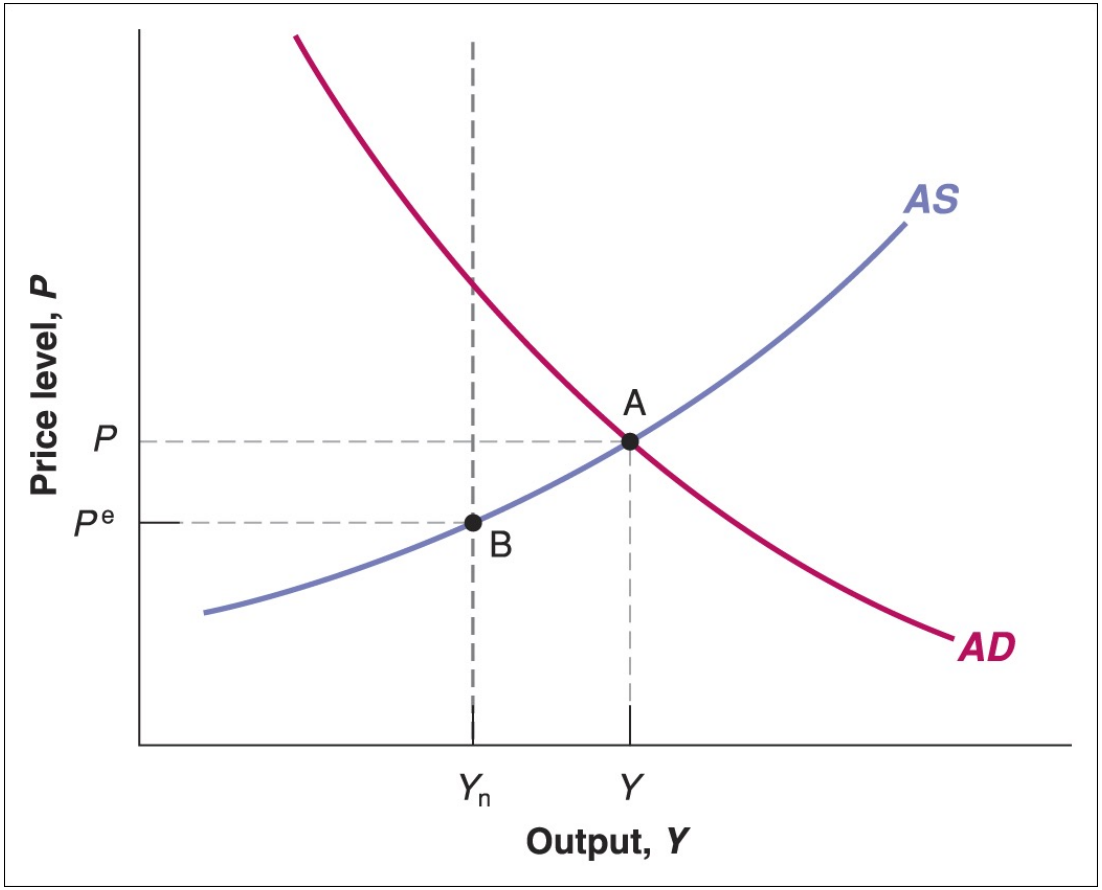
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Exercise 3: Is this economy in equilibrium in the short and the medium/long-run?

The equilibrium is given by the intersection of AD & AS curves.

At point *A*, the labor market, the goods market and financial market are all in equilibrium.

- *AS* is drawn for a given value of P^e. The higher the level of output, the higher the price level.
- *AD* is drawn for given values of *M, G* and *T*. The higher the price level, the lower the level of output.



The short-run equilibrium

Exercise 3: Is this economy in equilibrium in the short and the medium/long-run?

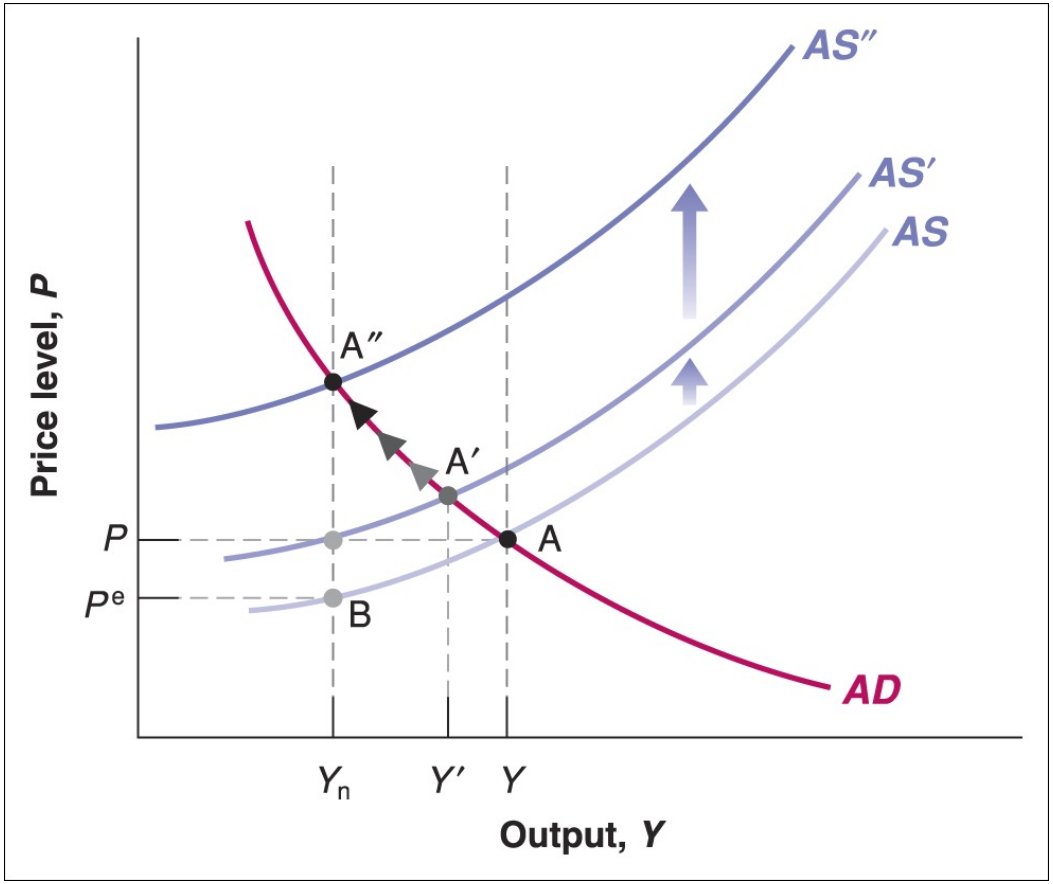
Ans: NO.

- At point *A*,

$$Y > Y_n \Rightarrow P > P^e$$

- Wage setters will upwardly revise their expectations of the future price level. This will cause the *AS* curve to shift upward.

- Expectation of a higher price level also leads to a higher nominal wage, which in turn leads to a higher price level.



The adjustment of output over time

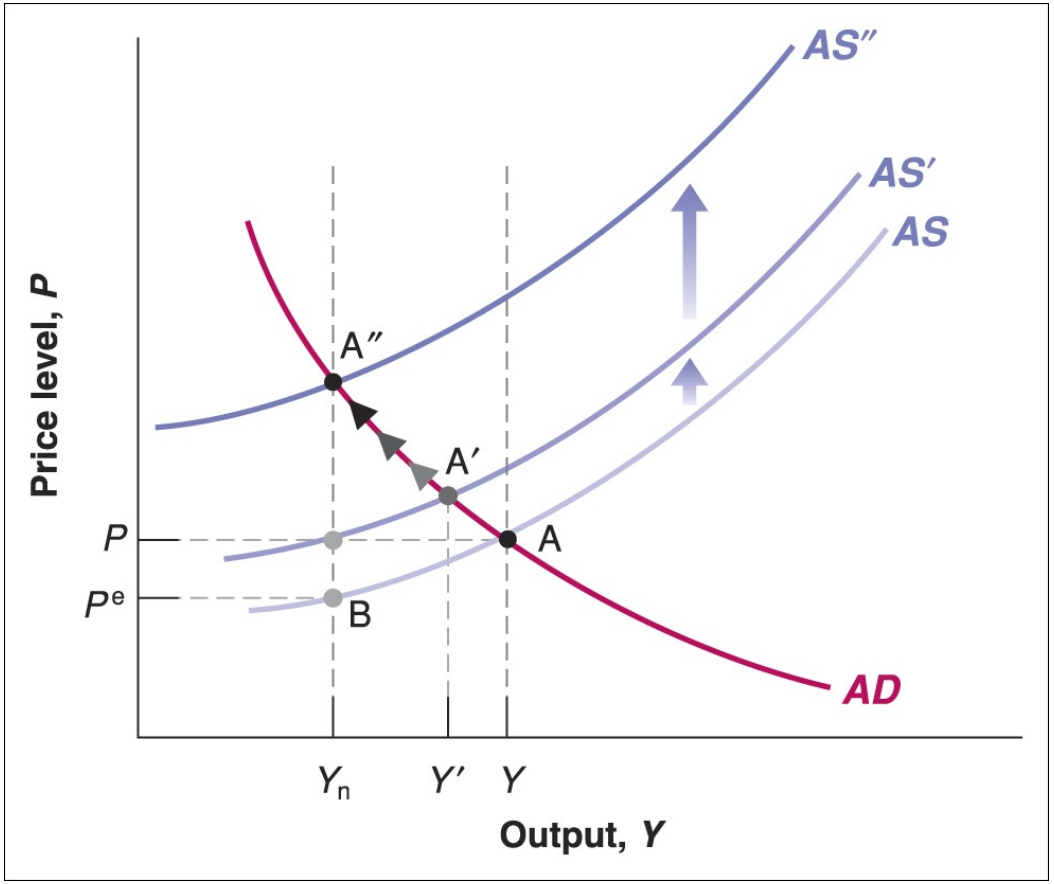
Exercise 3: Is this economy in equilibrium in the short and the medium/long-run?

If output is above the natural level of output, the *AS* curve shifts up over time until output has fallen back to the natural level of output.

$$Y = Y_n \text{ and } P = P^e$$

- The adjustment ends once wage setters no longer have a reason to change their expectations.

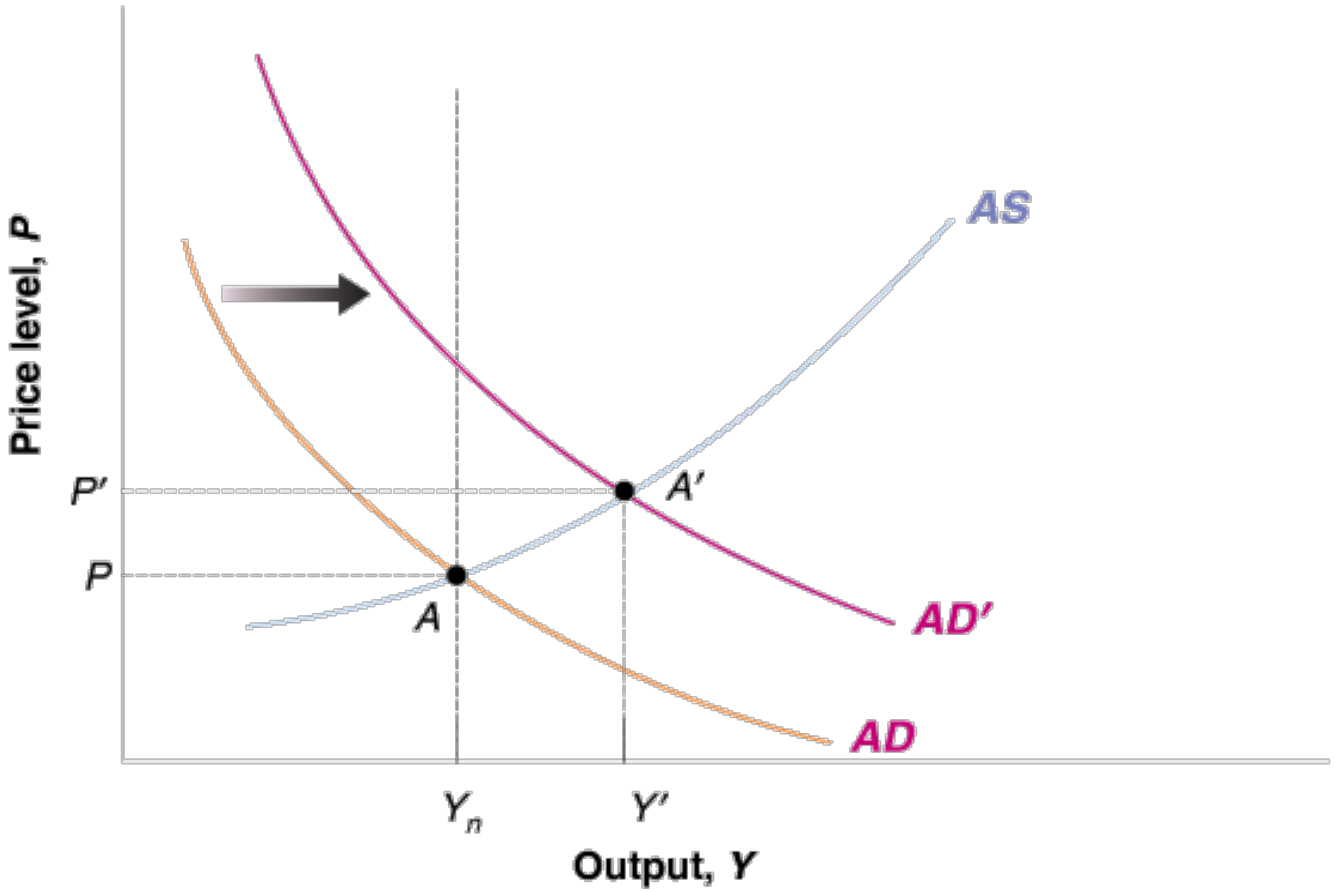
- In the medium run, output returns to the natural level of output.



The adjustment of output over time

Exercise 4: What is the effect of an Expansionary Monetary Policy?

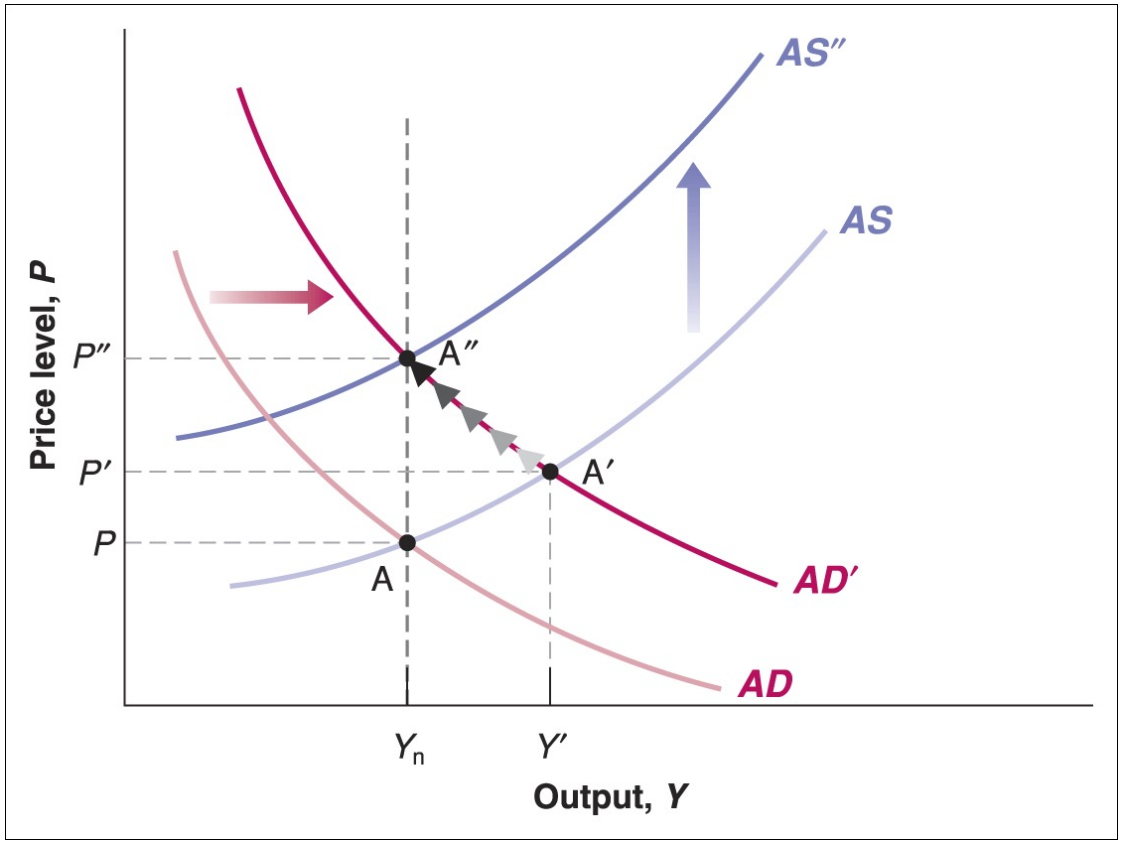
- The increase in the nominal Money stock causes AD curve to shift to the right.
- In the short run, output and the price level increase.



Exercise 4: What is the effect of an Expansionary Monetary Policy?

- The dynamics of adjustment

- The difference between Y and Y_n sets in motion the adjustment of price expectations.
- In the medium run, the *AS* curve shifts to *AS′* and the economy returns to equilibrium at Y_n.
- **The increase in prices is proportional to the increase in the nominal money stock: neutrality of money in the medium-run.**



The dynamic effects of a monetary expansion

A monetary expansion leads to an increase in output in the short run but has no effect on output in the medium run.

Exercise 4: What is the effect of an Expansionary Monetary Policy?

The neutrality of money

- In the *short run*, a monetary expansion leads to an increase in output, a decrease in the interest rate, and an increase in the price level.

- In the *medium run*, the increase in nominal money is reflected entirely in a proportional increase in the price level. The increase in nominal money has no effect on output or on the interest rate.

- **The neutrality of money in the medium run** does not mean that monetary policy cannot or should not be used to affect output.

Exercise 5: What is the effect of a Decrease in the Budget Deficit?

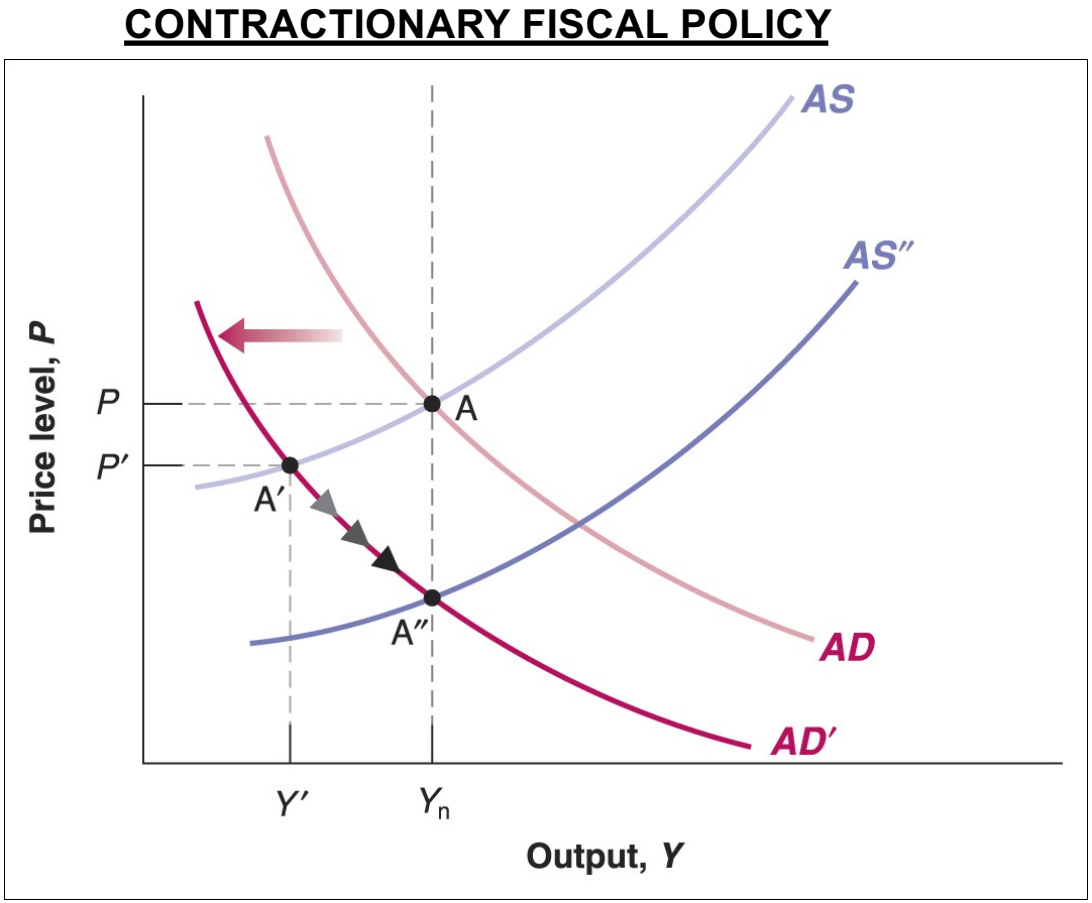
Imagine that the Government decides to decrease its Debt over GDP and, therefore, increases Income Taxes.

1. In our AD & AS Model, it can be represented as a movement of the AS curve to the left in the short-run. True or False?

2. How is the dynamic movement to the long-run equilibrium?

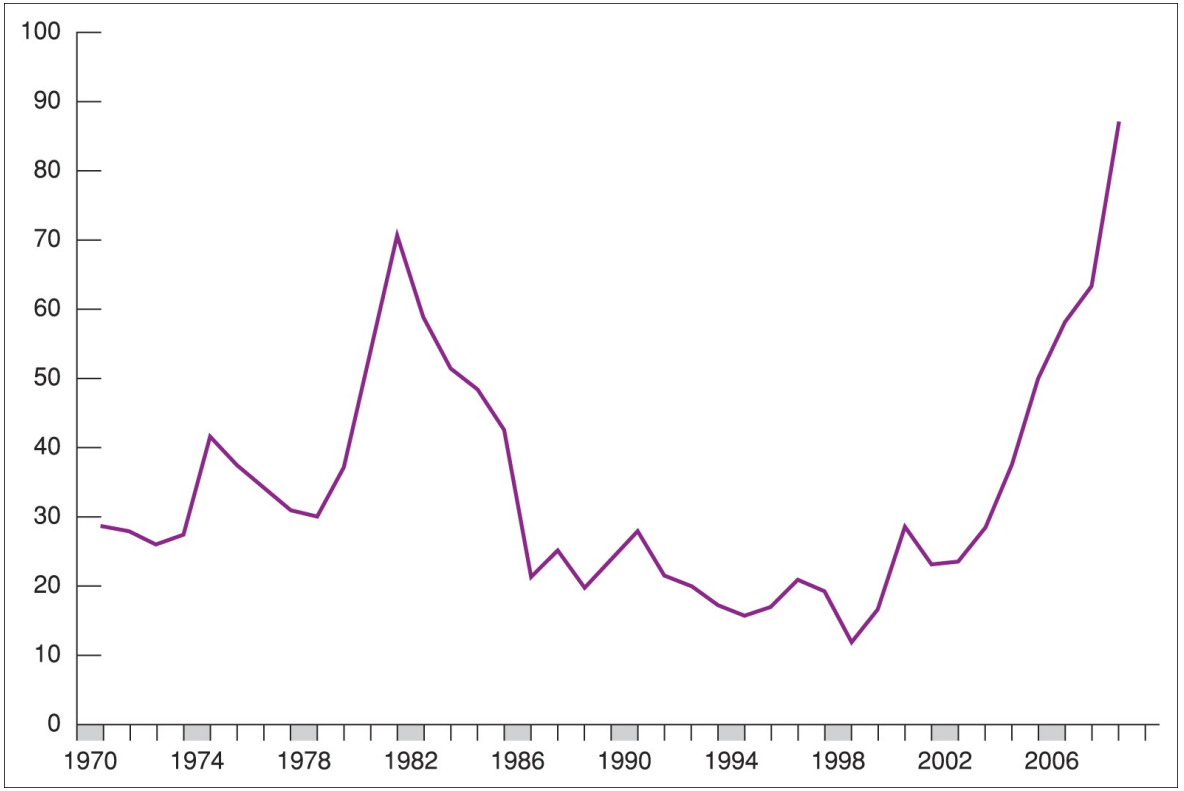
Exercise 5: What is the effect of a Decrease in the Budget Deficit?

- False. A decrease in the budget deficit leads initially to a decrease in output: AD shifts to the left.
- Over time, however, output returns to the natural level of output as AS shifts to the right.



Exercise 6: Which are the consequences of Changes in the Price of Oil?

Each of the two large price increases of the 1970s was associated with a sharp recession and a large increase in inflation—a combination macroeconomists call **stagflation**, to capture the combination of *stag*nation and in*flation* that characterized these episodes.



The real price of oil since 1970

There were two sharp increases in the relative price of oil in the 1970s, followed by a decrease until the 1990s, and a large increase since then.

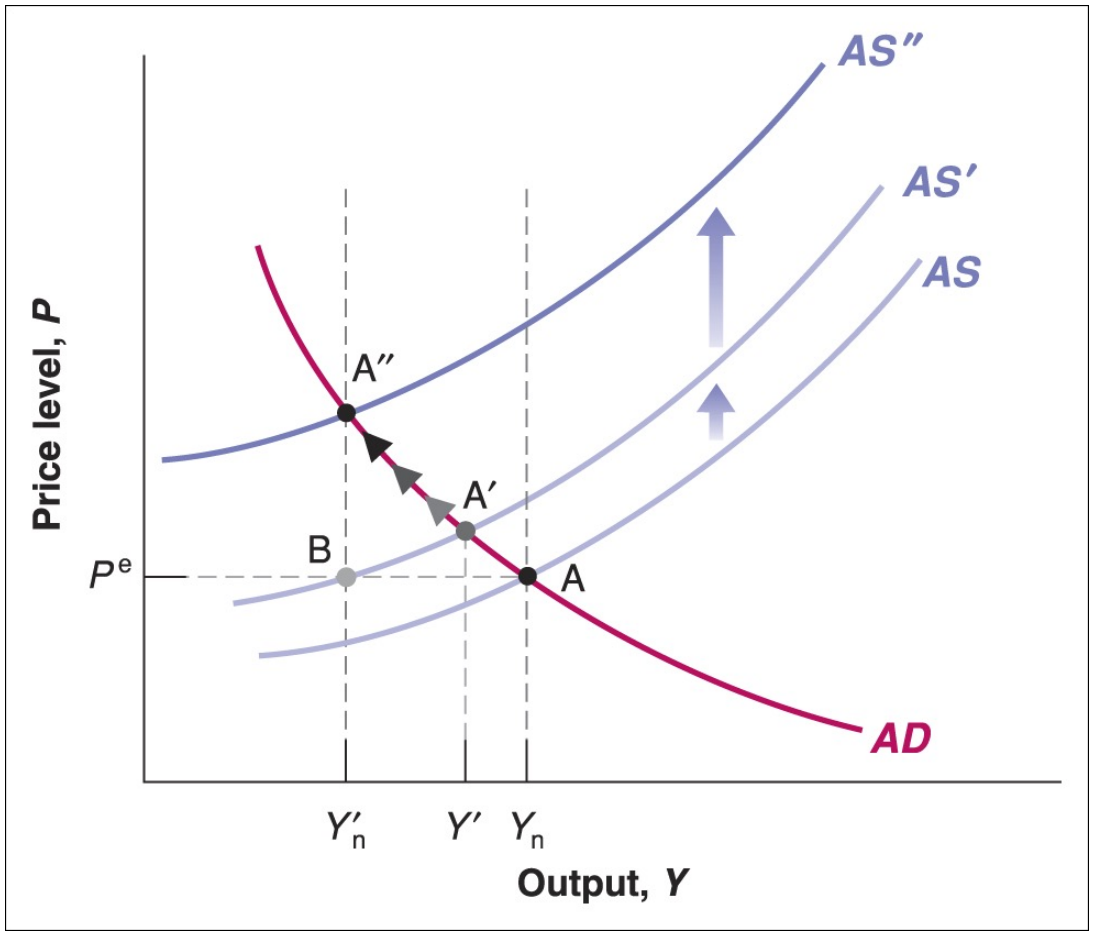
Source: Energy Information Administration (EIA) – Official Energy Statistics from the US Government, Eurostat.

Exercise 6: Which are the consequences of Changes in the Price of Oil?

The dynamics of adjustment

- After the increase in the price of oil, the new *AS* curve goes through point B, where output equals the new lower natural level of output, Y'_n, and the price level equals P^e.

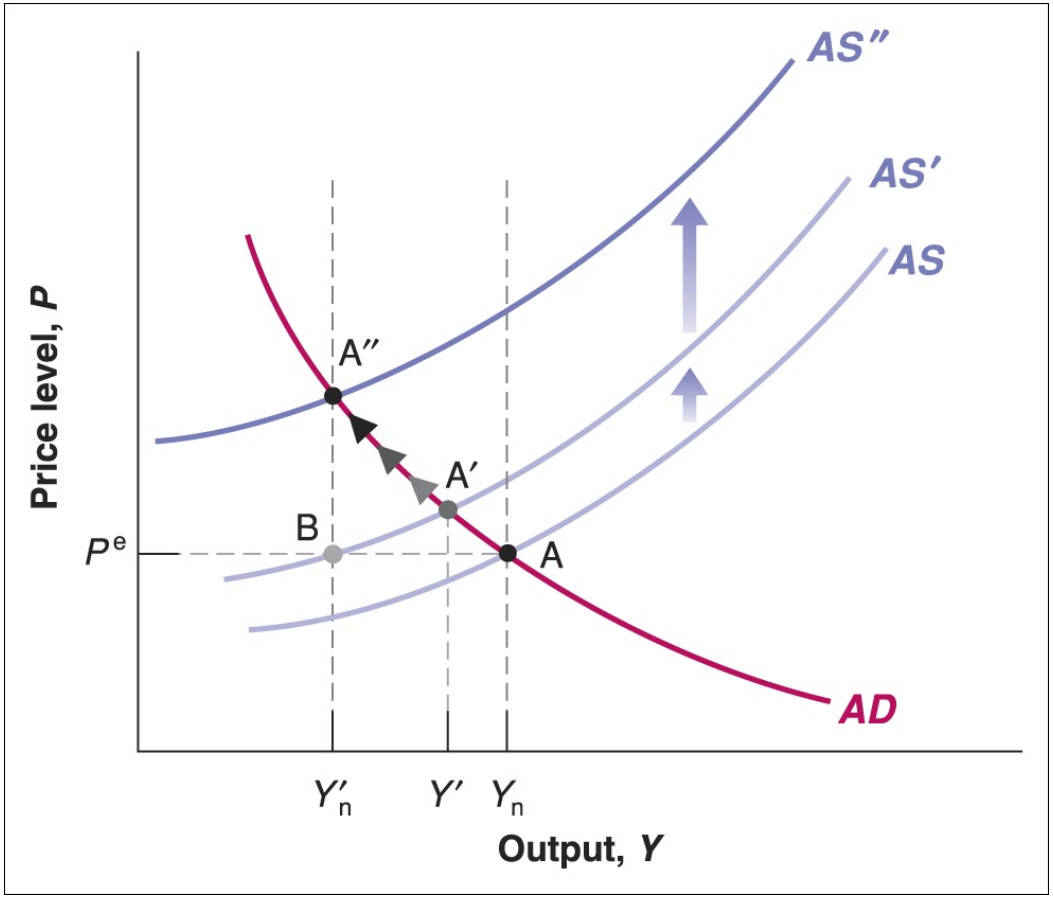
- The economy moves along the *AD* curve, from *A* to *A'*. Output decreases from Y_n to *Y'*.



Exercise 6: Which are the consequences of Changes in the Price of Oil?

An increase in the price of oil leads, in the short run, to a decrease in output and an increase in the price level.

Over time, output decreases further and the price level increases further: A¨



Exercise 7: Is this Table correct?

Short-run effects and medium-run effects of
a monetary expansion, a budget deficit reduction and an increase in oil price

	(Short run)			(Medium run)		
	Output level	**Interest rate**	**Price level**	**Output level**	**Interest rate**	**Price level**
Monetary expansion	**Increase**	**Decrease**	**Increase (small)**	**No change**	**No change**	**Increase**
Deficit reduction	**Decrease**	**Decrease**	**Decrease (small)**	**No change**	**Decrease**	**Decrease**
Increase in oil price	**Decrease**	**Increase**	**Increase**	**Decrease**	**Increase**	**Increase**

REFERENCES

Acemoglu Daron, David Laibson, and James List (2016), Macroeconomics, Global Edition (1st), Pearson.

Blanchard, Olivier, Alessia Amighini and Francesco Giavazzi (2017), Macroeconomics: A European Perspective, European Edition (2nd), Pearson.

Jones, Charles (2018), Macroeconomics, International Student Edition, W.W. Norton & Company.

Mankiw, N. Gregory (2018), Macroeconomics, International Edition (10th), Macmillan Learning.

Mankiw, N. Gregory (2020), Principles of Economics, 9th Edition, Cengage.